To be considered the problem, the **turn in** option should be checked.

Please upload for every problem **a single pdf** file named `Group_Pr_X_Last Name_First name{s}_CS_en_3Feb2021.pdf` where Group is the number of your group (ex 30311) and X is the number of problem. Please upload **only** in the appropriate assignement.

To be considered the problem, the **turn in** option should be checked.

Linear algebra, winter 2021, CS.

Problema 3

1. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear mapping represented, in canonical basis, by the matrix

$$M = \begin{pmatrix} 1 & 1 & -1 & 1 \\ 1 & 0 & -1 & 1 \\ 5 & 2 & -5 & 5 \end{pmatrix}.$$

   1. Find a basis in $\operatorname{Im} T$ and $\ker T$.
   2. Determine the dimension of $\ker T$, and $\operatorname{Im} T$.
   3. Determine an orthonormal basis in $(\ker T)^{\perp}$.

2. Let $(V, \langle , \rangle)$ an inner product space of dimension n, and $s = \{v_1, v_2, \dots v_k\}, k < n$ a family of linear indpendent vectors. How many sets of vectors can be obtained by using Gram-Schmidt orthonormalization process? .

Note. In the file submitted please write computations, explanations and everything needed for a better understanding of the problem.